

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	2 November 2025
Team ID	NM2025TMID05911
Project Name	Laptop Request Catalog Item
Maximum Marks	4 Marks

Technical Architecture:

The technical architecture utilizes the ServiceNow Request Fulfillment workflow/flow driven by the Service Catalog

Example:

A Catalog Item is submitted via the Service Portal, triggering a Flow Designer which uses a Script Include to check the user's role before creating an SCTASK for the Desktop Support team.



Guidelines:

Include all the processes (As an application logic / Technology Block)

Provide infrastructural demarcation (Local / Cloud)

Indicate external interfaces (third party API's etc.)

Indicate Data Storage components / services

Indicate interface to machine learning models (if applicable)

Table-1: Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	The portal through which the user submits and tracks the request.	Service Portal (HTML, CSS/SCSS, AngularJS/React)
2.	Form Definition	Defines the fields and structure for capturing user requirements (e.g., laptop type, peripherals).	Service Catalog Item & Variables
3.	Request Handling	The primary record that tracks the lifecycle of the specific item requested.	Requested Item (RITM) record on the sc_req_item table
4.	Automation Engine	The tool that manages sequential steps, branching logic, and task creation.	Flow Designer (Preferred) or Workflow Engine
5.	Server-Side Logic	Custom logic used for complex lookups, calculations, or integrations.	Script Include (JavaScript) & Business Rules
6.	Fulfillment Work	Specific, assignable pieces of work for fulfillment teams (e.g., Procurement, Desktop Support).	Catalog Task (SCTASK) record on the sc_task table
7.	Asset Tracking	The system of record for tracking the laptop as a provisioned asset.	CMDB (Configuration Item) and Asset Management

Table-2: Components & Technologies:

S.No	Component	Description	Technology
1.	Scalability	The platform's ability to handle a high volume of concurrent requests without performance degradation.	ServiceNow Cloud Infrastructure (Built-in multi-instance architecture)
2.	Integrity & Security	Ensuring data correctness and restricting access based on roles and permissions.	Access Control Lists (ACLs) and Client Scripts (for input validation)
3.	Maintainability	The ease of updating, troubleshooting, and extending the application post-deployment.	Low-Code Flow Designer and Scoped Application architecture
4.	Reporting & Metrics	The ability to track performance data, such as request fulfillment time and backlog.	Performance Analytics (PA) and Reporting Engine (using sc_req_item data)
5.	User Experience (UX)	Providing a modern, fast, and consistent interface for submitting and tracking the request.	Service Portal and Bootstrap Framework (Responsive Design)